

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Electrical Engineering

2. Subject Name: Digital Electronic Circuits L-T-P: 3-1-0

Credits: 4

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

A. Introduction to Digital Electronics

- Importance of digital electronics in the modern world
- Historical Background and evolution of digital VLSI
- Existing and Potential applications
- Key market players
- Overview of the course and key takeaways

B. Switching Algebra

- Definition of Boolean Algebra,
- Postulates and Theorems of Boolean Algebra,
- Switching Functions (SOP and POS forms),
- Maxterms and Minterms, Incompletely Specified Functions

C. Simplification of Switching Functions

- Karnaugh Map Method (both SOP and POS forms),
- Quine-McCluskey Tabular Method

D. Different Logic Families: Digital Gates using CMOS and pass transistor logic

- Evolution of digital VLSI technology
- CMOS for digital circuits
- CMOS logic gates and pass transistor
- Power vs speed in CMOS – Propagation delay vs dynamic power

E. Combinational Logic Circuits

- Decoder, Encoder, Magnitude Comparators,
- Design with Mux, Arithmetic and logic elements
- Adder, subtractor, Arithmetic Logic Unit or ALU,

F. Sequential Logic Circuits

- SR, JK, D and T Flip-flops, shift registers and counters

G. Finite State Machines

- State transition table, design examples, Mealy and Moore machine models, ASM chart

H. Memory Devices

- ROM, RAM – SRAM and DRAM

I. Implementation of Digital Module Design with Some Case Studies [8 hours]

- Timing and data hazard in digital circuit design
- Case study: 4/8-bit ALU or others
- Register Transfer Logic
- Introduction to hardware descriptive language – either Verilog or VHDL
- HDL based implementation case study

Reference Books:

1. **Digital Design** by M. Morris Mano and Michael D. Ciletti
2. **Digital Integrated Circuits** by Jan M. Rabaey, Anantha Chandrakasan and Borivoje Nikolic
3. **Foundation of Analog and Digital Electronic Circuits** by Anant Agarwal and Jeffrey Lang
4. **Digital Fundamentals** by Thomas L. Floyd

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to Digital Electronics	2
2.	Switching Algebra	6
3.	Simplification of Switching Functions	4
4.	Different Logic Families	6
5.	Combinational Logic Circuits	6
6.	Sequential Logic Circuits	6
7.	Finite State Machines	4
8.	Memory Devices	2
9.	Implementation of Digital Module Design with Some Case Studies	8
Total		44